

# What to expect from Dart & Flutter on RISC-V

Fluttercon - Jul 2023



It's 2028, your Flutter app is in the top 10  
Huawei's latest foldable flagship has a RISC-V SoC





# Hi, I'm Chris

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<https://chris.swanz.net>



# Agenda

- What is RISC-V
- Why?
- Can I run Dart on my soldering iron?
- Dart on RISC-V today
- The road ahead
- What about Apple and cloud providers?



What is RISC-V?



RISC-V is an open standard  
Instruction Set Architecture (ISA)  
enabling a new era of processor  
innovation through open  
collaboration

<https://riscv.org/about/>





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Why?



EXECUTIVE ORDERS

# Executive Order on Securing the Information and Communications Technology and Services Supply Chain

— INFRASTRUCTURE & TECHNOLOGY

| Issued on: May 15, 2019



# RISC-V business: Tech foundation moving to Switzerland because of geopolitical concerns

Unanimous decision of board to up sticks from Delaware

 Tim Anderson

Tue 26 Nov 2019 // 11:35 UTC

The RISC-V Foundation, which directs the development of an open-source instruction set architecture for CPUs, will incorporate in Switzerland. Currently it is a non-stock corporation in Delaware, USA.

[https://www.theregister.com/2019/11/26/riscv\\_foundation\\_switzerland/](https://www.theregister.com/2019/11/26/riscv_foundation_switzerland/)



# Huawei running out of smartphone CPUs as US sanctions begin to bite

Chinese giant suggests current Kirin SoC could be last of its kind

 Robbie Harb

Mon 10 Aug 2020 // 05:02 UTC

Huawei will halt production of its flagship Kirin chipsets later this year, due to US sanctions on Chinese companies.

Production of the Chinese firm's Kirin and AI-powered chips, both of which are used to power its handsets, will be forced to stop on September 15 as a result of US sanctions on its suppliers, said Richard Yu, president of the company's consumer unit, at an event on Friday.



[https://www.theregister.com/2020/08/10/huawei\\_halts\\_chip\\_production\\_as/](https://www.theregister.com/2020/08/10/huawei_halts_chip_production_as/)



# Western Digital Transitions to RISC-V Open-Source Architecture for Big Data, IoT



Western Digital believes open-source architecture is the best solution for Big Data and Fast Data and has committed to transition all of its cores to RISC-V.

Chris Wiltz | Dec 05, 2017

<https://www.designnews.com/electronics-test/western-digital-transitions-risc-v-open-source-architecture-big-data-iot>



“We're not doing this for cost reasons. This is all completely and totally about innovation. ... Unlocking our ability to bring innovation to data and process it in ways we can't contemplate today.”

Martin Fink, CTO, Western Digital





Can I run Dart on my soldering iron?

# PINECIL – Smart Mini Portable Soldering Iron



|                 |                                                                                                                                                                                  |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PLATFORM</b> | <ul style="list-style-type: none"><li>• OS: Ralim's IronOS build</li><li>• Chipset: Bouffalo BL-706</li><li>• CPU: 32-bit RV32IMAFC RISC-V "SiFive E24 Core" @ 144 MHz</li></ul> |
| <b>DISPLAY</b>  | <ul style="list-style-type: none"><li>• Type: OLED White Colour Monochrome Display</li><li>• Size: 0.69 inches</li><li>• Resolution: 96x16pixels</li></ul>                       |
| <b>MEMORY</b>   | <ul style="list-style-type: none"><li>• Internal Flash Memory: 192KB</li><li>• System Memory: 132KB SRAM</li><li>• Expansion: None</li></ul>                                     |



# MCU vs SoC

## RISC-V MCUs:

- 32 or 64 bit
- RTOS
- Toothbrushes, clocks
- Low power IoT devices

- Can't run Dart

## RISC-V SoCs:

- 64 bit
- Linux
- Phones, tablets, set top boxes
- High power IoT devices

- Can run Dart



# Dart is a dynamically linked language.

## Let's take a peek inside /runtime:

```
# riscv64 - ignoring certs stuff

./etc:
-rw-r--r-- 1 root root 494 Sep 22 2022 nsswitch.conf

./lib:
-rwxr-xr-x 2 root root 124920 Apr 10 08:35 ld-linux-riscv64-lp64d.so.1

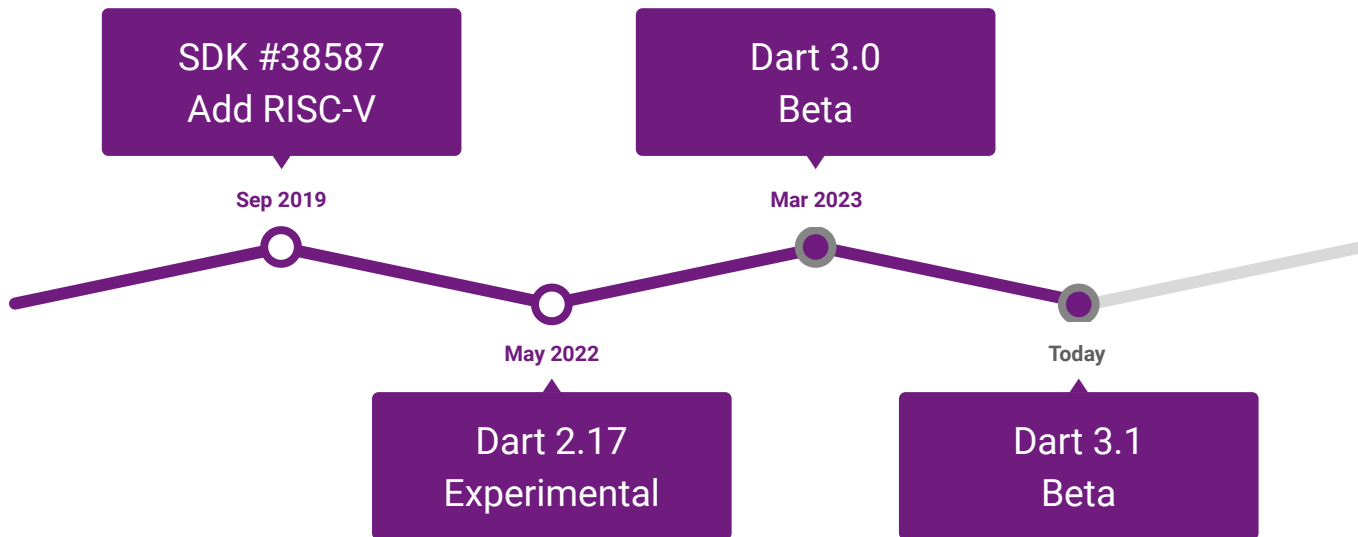
./usr/lib/riscv64-linux-gnu:
-rw-r--r-- 1 root root 22480 Jun 11 11:48 libatomic.so.1

./lib/riscv64-linux-gnu:
-rw-r--r-- 1 root root 6160 Apr 10 08:35 librt.so.1
-rw-r--r-- 1 root root 43384 Apr 10 08:35 libresolv.so.2
-rw-r--r-- 1 root root 5856 Apr 10 08:35 libpthread.so.0
-rw-r--r-- 1 root root 5856 Apr 10 08:35 libnss_dns.so.2
-rw-r--r-- 1 root root 440768 Apr 10 08:35 libm.so.6
-rw-r--r-- 1 root root 5856 Apr 10 08:35 libdl.so.2
-rwxr-xr-x 1 root root 1213544 Apr 10 08:35 libc.so.6
-rwxr-xr-x 2 root root 124920 Apr 10 08:35 ld-linux-riscv64-lp64d.so.1
```

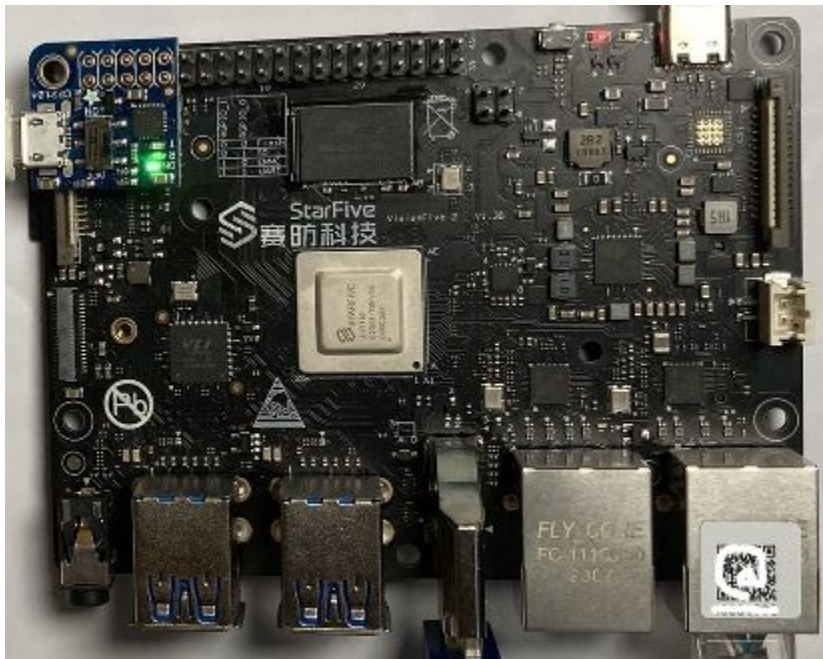


Dart on RISC-V today

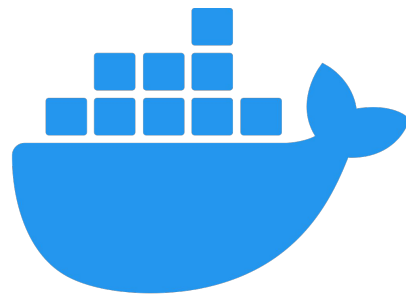
# The story so far



# Dart on real RISC-V hardware



```
# dart --version  
Dart SDK version: 3.1.0-163.1.beta (beta)  
(Wed Jun 7 05:06:41 2023 -0700) on "linux_riscv64"
```



docker®





AMPUTATING ARM —

# Google wants RISC-V to be a “tier-1” Android architecture

Google's keynote at the RISC-V Summit promises official, polished support.

RON AMADEO - 1/3/2023, 9:14 PM

<https://arstechnica.com/gadgets/2023/01/google-announces-official-android-support-for-risc-v/>



# Lars Bergstrom



<https://www.youtube.com/watch?v=xLwdUn3DQp8>

# Progress in 2023: AOSP

Android Runtime (ART) available

Cuttlefish emulator available

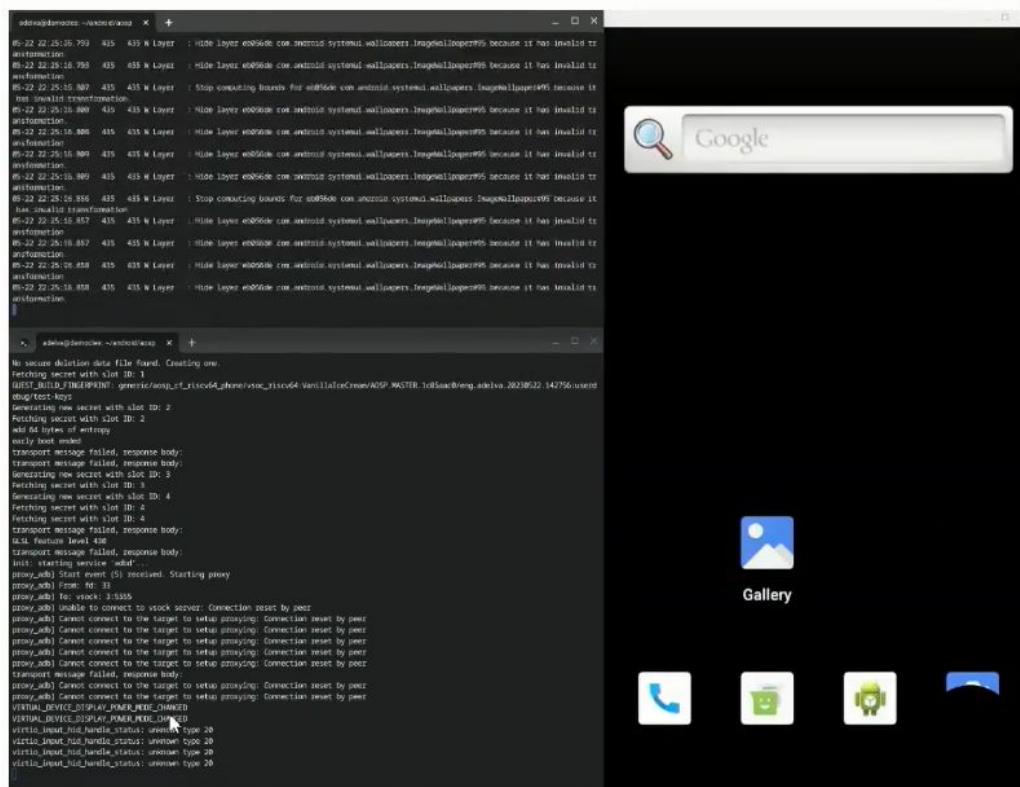
Prebuild tools - compilers & system root libraries available

Initial support landing soon for vector & bit manipulation extensions beyond RV64GC to optimise the platform

Profiling works (prebuilts coming soon)



# Cuttlefish RISC-V Emulator



<https://github.com/google/android-riscv64>

\$ lunch aosp\_cf\_riscv64\_phone-userdebug

\$ m -j

\$ launch\_cvd -cpus=8 -memory\_mb=8192

Then, use vncviewer to connect



# Open Source Projects

Contributions and work with ecosystem partners and upstream maintainers on:

- LLVM
- Kernel
- QEMU
- Graphics libraries
- Crypto libraries
- Codecs



# Upstream with RISC-V

Work to ensure ABI is forwards compatible with new potential atomics additions

Support for many RISC-V members on the Android SIG looking into standing up AOSP on a variety of emulation and physical devices



## Languages - Dart on RISC-V

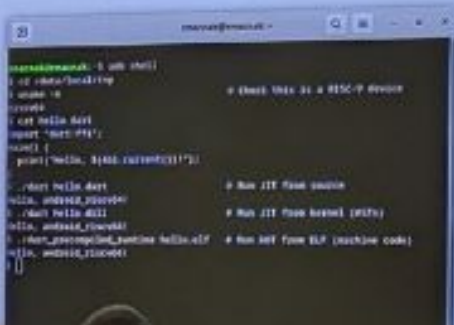
Dart: fast apps on any platform

Dart, with Flutter, powers more than 1M apps in Google Play, e.g.

- Alibaba, BMW, ByteDance, eBay, Google, Tencent, ...

Dart is

- Memory-safe, garbage-collected
- JIT for dev, AOT for prod: ARM, x86
- Experimental RISC-V on android/riscv64



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[illegible]

The road ahead

# Android Roadmap

## 2023 Q3

Virtual devices with  
accelerated graphics

Android Runtime (ART)  
optimisations

Optimisations landing  
for QEMU, kernel and  
all in tree libraries



## 2023 Q4

NDK ABI finalised  
and canary builds  
available on public CI

RISC-V on x86-64  
and Arm64 for easier  
testing

## 2024

Emulators available  
with full feature set  
to test apps on  
various device form  
factors

Release NDK with  
RISC-V support



# Android Profiles

Supported ABI will be added to the CDD list as riscv64 (no 32bit)

All “supported instruction sets” will be a **combination** of:

- A RISC-V profile (probably RVA22)
- Ratified extensions (probably vector + vector crypto)
- Intentional omissions: SIMD, Scalar Crypto

Will require Android compatible devices to be **conforming** hardware

- Must correctly implement the RISC-V ISA
- Must not misuse elements of the encoding space reserved for future extensions

 Platforms (but not applications) can take advantage of the RISC-V features in the reserved vendor space



# Full Stack Dart needs Debian



| Release      | on           | called   | RISC-V |
|--------------|--------------|----------|--------|
| 11           | Aug 2021     | Bullseye |        |
| 12 (stable)  | Jun 2023     | Bookworm |        |
| 13 (testing) | Summer 2025? | Trixie   |        |
| Unstable     | Rolling      | Sid      | ✓      |





**Goal:** Accelerate open source SW for RISC-V architecture

**How:** Align on highest priorities & avoid (accidental) duplication of work



# Focus Areas



**Coordination and collaboration among the RISE members is across an array of software areas to deliver high quality and high performance implementations for RISC-V software.**

|                          |                                                    |
|--------------------------|----------------------------------------------------|
| Compilers & Toolchains   | LLVM, GCC                                          |
| System Libraries         | Glibc, OpenSSL, OpenBLAS, LAPACK, OneDAL, Jemalloc |
| Kernel & Virtualization  | Linux, Android                                     |
| Language Runtimes        | Python, OpenJDK/Java, V8                           |
| Linux Distro Integration | Ubuntu, Debian, RHEL, Fedora, Alpine               |
| Debug & Profiling Tools  | Performance profiles, DynamoRIO, Valgrind          |
| Simulator/Emulators      | QEMU, SPIKE                                        |
| System Software          | UEFI, ACPI                                         |



What about Apple?

What about cloud?

# Review

- RISC-V is an open standard Instruction Set Architecture (ISA)
- Geopolitics, innovation, and cost are driving adoption
- You can't Dart on a PINECIL soldering iron
- Dart 3.x betas run great on RISC-V dev boards today
- Some big chunks of infrastructure aren't ready yet
  - ◆ Looks like >2y but <5y work from here
- @Apple & cloud provider secrecy makes them hard to predict



Call to action: Try running Dart on RISC-V in QEMU



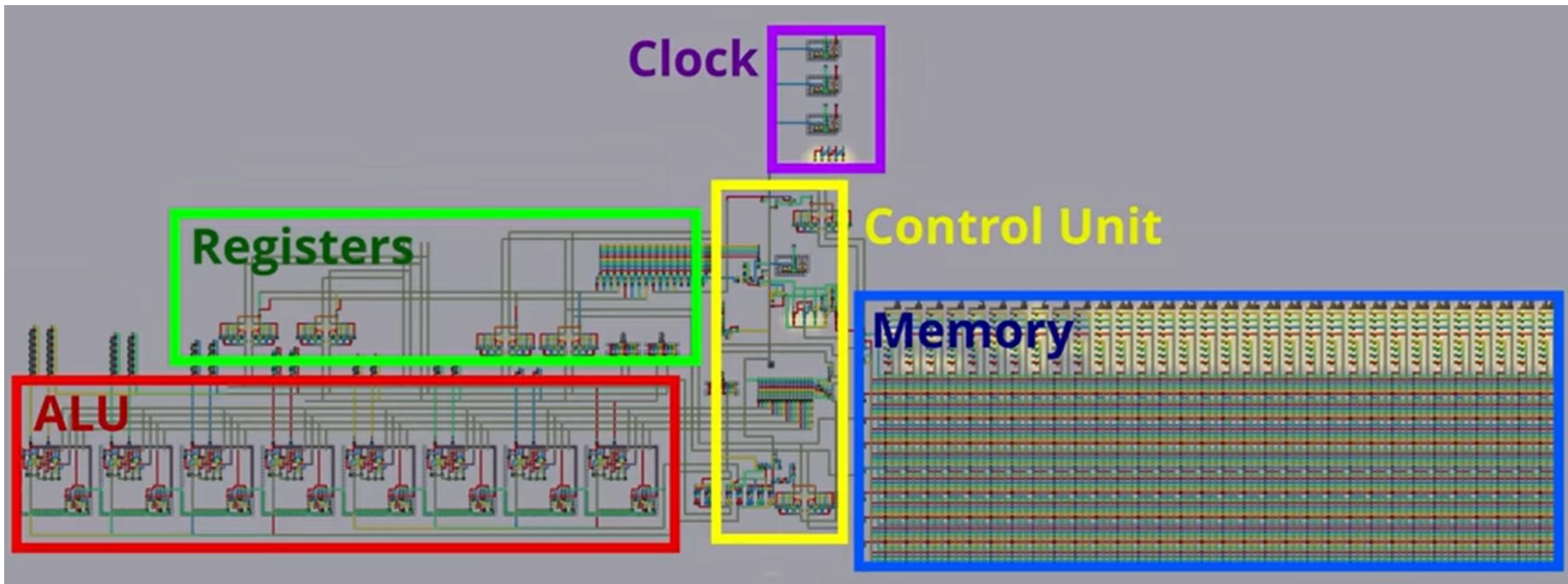
<https://wiki.ubuntu.com/RISC-V/QEMU>



**QEMU**



# RISC-V computer inside Terraria



<https://www.pcgamer.com/wunderkind-builds-fully-compliant-risc-v-computer-entirely-inside-terraria-then-plays-pong-on-it/>



# Resources

## Blog posts

<https://blog.thestateofme.com/2017/12/01/risc-v-innovation-and-not-having-to-ask-permission/>

<https://blog.thestateofme.com/2019/05/22/open-source-and-export-controls/>

## Bookmarks

<https://pinboard.in/u:cpswan/t:RISC-V>

## RISC-V International on YouTube

 <https://www.youtube.com/@RISCVInternational/videos>



# Thanks for your time

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Questions?